



# Effects of nitrogen addition on soil methane uptake in global forest biomes<sup>☆</sup>

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## ABSTRACT

Nitrogen (N) deposition has been conventionally thought to decrease forest soil methane (CH<sub>4</sub>) uptake, while the biome specific and dose dependent effect is poorly understood. Based on a meta-analysis of 63 N addition trials from 7 boreal forests, 8 temperate forests, 13 subtropical and 4 tropical forests, we evaluated the effects of N addition on soil CH<sub>4</sub> uptake fluxes across global forest biomes. When combining all N addition levels, soil CH<sub>4</sub> uptake was insignificantly decreased by 7% in boreal forests, while N addition significantly decreased soil CH<sub>4</sub> uptake by 39% in temperate forests and by 21% in subtropical and tropical forests, respectively. Meta-regression analyses, however, indicated a shift from a positive to a negative effect on soil CH<sub>4</sub> uptake with increasing N additions both in boreal forests (threshold = 48 kg N ha<sup>-1</sup> yr<sup>-1</sup>) and temperate forests (threshold = 27 kg N ha<sup>-1</sup> yr<sup>-1</sup>), while no such shift was found in subtropical and tropical forests. Considering that current N deposition to most boreal and temperate forests is below the abovementioned thresholds, N deposition likely exerts a positive to neutral effect on soil CH<sub>4</sub> uptake in both forest biomes. Our results provide new insights on the biome specific and dose dependent effect of N addition on soil CH<sub>4</sub> sink in global forests and suggest that the current understanding that N deposition decreases forest soil CH<sub>4</sub> uptake is flawed by high levels of experimental N addition.

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## 1. Introduction

Concentrations of atmosphere methane (CH<sub>4</sub>) have been growing rapidly over recent decades and aroused increasing concerns about its effect on climate warming (Dlugokencky et al., 2011; Kirschke et al., 2013). Compared to other land uses, forest soils are an important sink of atmospheric CH<sub>4</sub> that dominates the total soil CH<sub>4</sub> uptake in terrestrial biomes (Le Mer and Roger, 2001; Dutaer and Verchot, 2007). Anthropogenic nitrogen (N) emissions have increased N deposition to global forests (Lamarque et al., 2013; Vet et al., 2014; Schwede et al., 2018), resulting in various ecological consequences (De Vries et al., 2017; Du et al., 2019; Gilliam et al., 2019; Schmitz et al., 2019; Du et al., 2020a). Current

understanding of the effects of N deposition on forest soil CH<sub>4</sub> uptake is mainly derived from N addition experiments. Although forest biomes generally receive relatively low levels of N deposition (predominantly < 50 kg N ha<sup>-1</sup> yr<sup>-1</sup>; Vet et al., 2014; Schwede et al., 2018), high N dosages (e.g., > 100 kg N ha<sup>-1</sup> yr<sup>-1</sup>) have been frequently used in existing N addition experiments. This leads to limited insights on soil CH<sub>4</sub> flux in response to N deposition in global forests (Liu and Greaver, 2009; Aronson and Helliker, 2010; Bodelier and Steenbergh, 2014; De Vries et al., 2017).

Soil CH<sub>4</sub> sink results from a higher rate of CH<sub>4</sub> oxidation by methanotrophs than that of methanogenesis by methanogens, both of which are sensitive to external N additions (Schnell and King, 1994; Le Mer and Roger, 2001; Bodelier and Laanbroek, 2004). In an N-deficient ecosystem, N inputs might release the N limitation of methanotrophic microorganisms and/or the biosynthesis of enzymes involved in methane oxidation and thus benefit soil CH<sub>4</sub> uptake (Bodelier et al., 2000; Bodelier and Laanbroek, 2004; Reay and Nedwell, 2004). In addition, external N inputs can stimulate forest growth (Sonnleitner et al., 2001; Du and De Vries, 2018;

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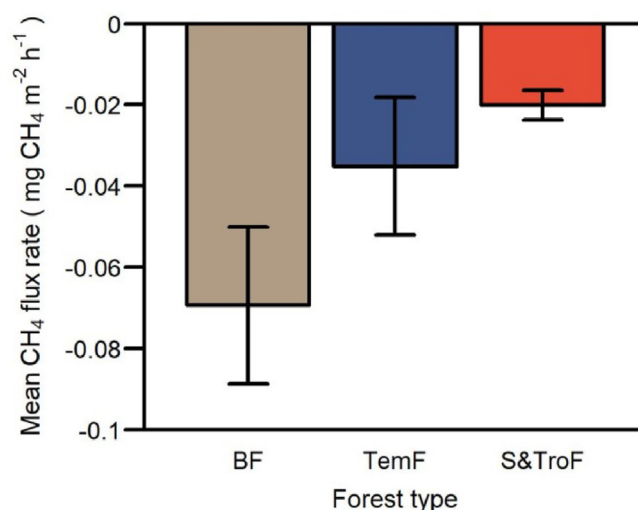
Schulte-Uebbing and De Vries, 2018) and increase evapotranspiration, indirectly favoring microbial CH<sub>4</sub> oxidation in soils. However, excessive N inputs may decrease soil CH<sub>4</sub> uptake via (i) a direct inhibition of CH<sub>4</sub> oxidation due to either osmotic stress caused by increased soil inorganic N concentrations or competitive inhibition of the enzyme methane mono-oxygenase by ammonia (Schnell and King, 1994; Sitaula et al., 1995; Bodelier and Laanbroek, 2004) and (ii) an indirect effect due to soil acidification and an imbalance of N and phosphorus (Veraart et al., 2015; Zhang et al., 2020). However, the potential dose dependent effect has not been investigated in previous meta-analyses (Liu and Greaver, 2009; Aronson and Helliiker, 2010).

Previous studies have indicated a poleward increase in N limitation from tropical forest towards boreal forest, both internally driven by mineralization and externally from biological N fixation and N deposition (Cleveland et al., 2013; Du and De Vries, 2018; Deng et al., 2018; Du et al., 2020b). This may imply a biome-specific response of soil CH<sub>4</sub> fluxes to N addition due to a shift in background N availability. Based on a synthesis of experimental results in literature, we evaluated the effects of N addition on soil CH<sub>4</sub> fluxes across boreal, temperate, subtropical and tropical forests. We hypothesize that 1) the effect of N addition on soil CH<sub>4</sub> uptake varies with the level of N addition and 2) differs between forest biomes with relatively low-level N availability (e.g., boreal and temperate forests) and those with high-level N availability (e.g., subtropical and tropical forests).

## 2. Data and method

### 2.1. Data set

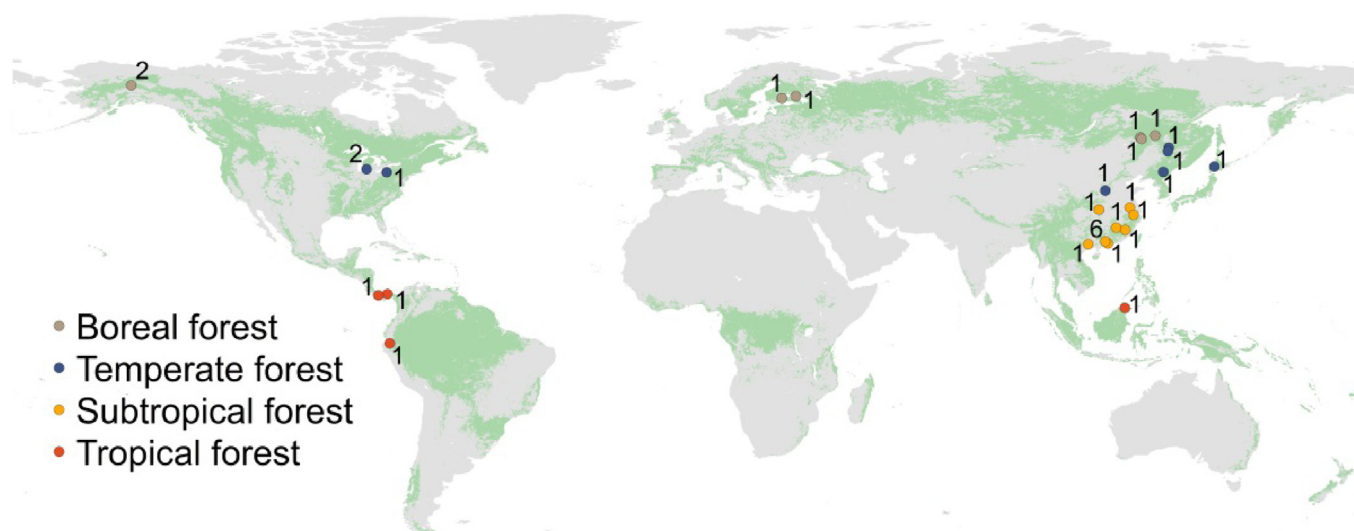
By conducting a survey on the online library of ISI Web of Science (<http://isiknowledge.com>), Google Scholar (<https://scholar.google.com>) and China National Knowledge Infrastructure (<http://www.cnki.net/>), we collected field experimental data on the effects of N addition on soil CH<sub>4</sub> fluxes in forest ecosystems across the globe. The key words ‘methane (or CH<sub>4</sub>)’, ‘nitrogen addition (or nitrogen fertilization, nitrogen deposition)’ and ‘forest’ were used. We recorded data on the means and standard deviations of soil CH<sub>4</sub> flux in control plots and treatment plots, as well as information on experimental treatments (the number of replicates, N addition rate,



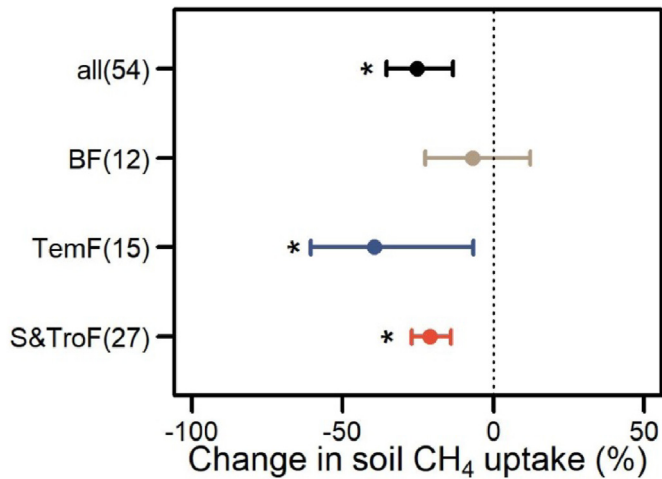
**Fig. 2.** Mean soil CH<sub>4</sub> flux (mg CH<sub>4</sub> m<sup>-2</sup> h<sup>-1</sup>) in boreal forests (BF, n = 7), temperate forests (TemF, n = 8) as well as subtropical and tropical forests (S&TroF, n = 17). Values for boreal and temperate forests are growing-season means, while the annual mean value is shown for subtropical and tropical forests. Error bars are indicated using standard error.

the duration of the treatment, N forms [i.e., NO<sub>3</sub><sup>-</sup>-N, NH<sub>4</sub><sup>+</sup>-N, NH<sub>4</sub>NO<sub>3</sub>—N or urea]) and environmental factors (site location [latitude and longitude], climate [i.e., mean annual temperature, MAT, and mean annual precipitation, MAP ], soil pH and forest type [i.e., boreal, temperate, subtropical or tropical forests]).

Values were directly used from tables or digitized from figures using a GetData Graph Digitizer (Version 2.25, <http://www.getdatagraph-digitizer.com>). When data on soil pH were not reported in the literature, we retrieved the values from SoilGrids ([www.soilgrids.org](http://www.soilgrids.org)) according to the location of the study sites. Negative values of soil CH<sub>4</sub> flux indicate an uptake of CH<sub>4</sub> by soils, while positive values indicate an emission of CH<sub>4</sub> from soils. In boreal and temperate forests, soil CH<sub>4</sub> fluxes were generally measured during the growing season, while measurements were conducted whole year round in subtropical and tropical forests (see more details in Table S1).



**Fig. 1.** Geographical distribution of nitrogen addition experiments from 32 forests across 26 sites. The numbers of experimental studies at each site are shown. Green shadows indicate the distribution of global forest. (For interpretation of the references to colour in this figure legend, the reader is referred to the Web version of this article.)



**Fig. 3.** Relative changes (%) in soil CH<sub>4</sub> uptake in response to N additions in boreal (BF), temperate (TemF) as well as subtropical and tropical forests (S&TroF). The asterisk (\*) indicates a significant difference from zero ( $p < 0.05$ ).

Our database includes experimental results of 63 -N addition trials across 32 forests from 26 sites (Fig. 1). Specifically, 44 trials were conducted by using  $\text{NH}_4\text{NO}_3$ , 15 trials by using urea, and 4 paired trials by using both nitrate ( $\text{NaNO}_3$  or  $\text{KNO}_3$ ) and ammonium ( $\text{NH}_4\text{Cl}$ ) based N forms. For the paired trails, the rates of soil CH<sub>4</sub> uptake were averaged for each addition of reduced and oxidized N forms. Finally, there were 12, 22, 25, and 4 fertilization trials from 7 boreal forests, 8 temperate forests, 13 subtropical forests and 4 tropical forests, respectively (Table S1). Data of tropical forests were aggregated with those of subtropical forests for further analysis

because there were too few trials for a robust statistical analysis ( $n = 4$ ).

## 2.2. Statistical analysis

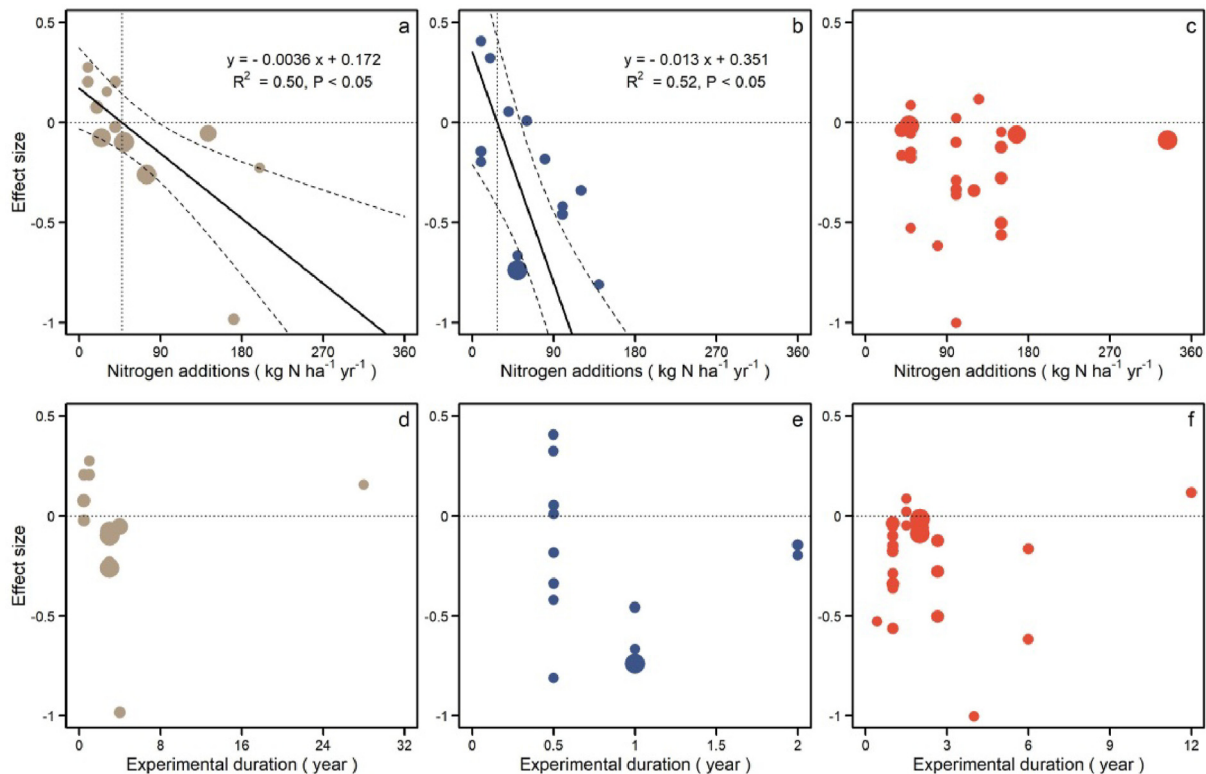
To quantify the impacts of N addition soil CH<sub>4</sub> fluxes, we defined an effect size as given in equation (1) (Hedges et al., 1999),

$$\text{effect size} = \ln \frac{\text{flux}_t}{\text{flux}_c} = \ln(\text{flux}_t) - \ln(\text{flux}_c) \quad (1)$$

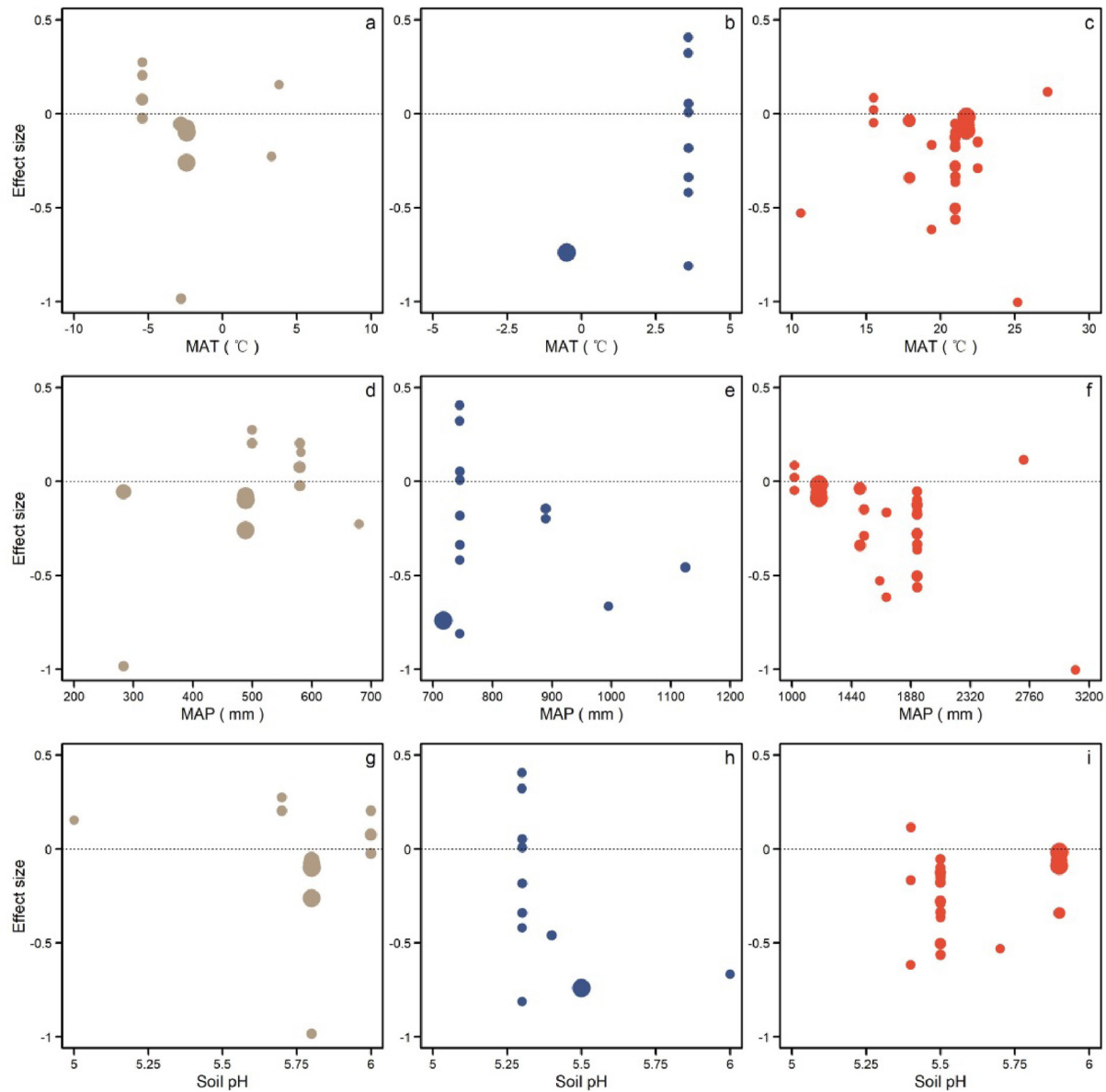
where  $\text{flux}_t$  and  $\text{flux}_c$  indicate the mean soil CH<sub>4</sub> fluxes in the N treatment plots and control plots ( $\text{mg CH}_4 \text{ m}^{-2} \text{ h}^{-1}$ ), respectively. We had to exclude nine N addition trails from two temperate forests and two tropical forests (see Table S1) because the negative ratios of  $\text{flux}_t$  versus  $\text{flux}_c$  hindered an estimate of the above-mentioned effect size. The variance ( $v_i$ ) of the effect size was calculated based on the mean soil CH<sub>4</sub> fluxes, standard deviations ( $sd_t$  and  $sd_c$ ) and number of replicates ( $n_t$  and  $n_c$ ) for the N treatment and control plots, according to equation (2) as given below,

$$v_i = \frac{(sd_t)^2}{n_t(\text{flux}_t)^2} + \frac{(sd_c)^2}{n_c(\text{flux}_c)^2} \quad (2)$$

A random effect model was used to estimate mean effect size of N additions (Borenstein et al., 2009), being calculated as the mean of individual effect size with weights ( $w_i$ ) equal to the inverse of the sum of the within-study variance ( $v_i$ , due to sampling error) and between-study variance ( $\tau^2$ , due to variation in experimental conditions) (equation (3)),



**Fig. 4.** Meta-regression plots showing changes of effect size with N additions ( $\text{kg N ha}^{-1} \text{ yr}^{-1}$ ) and experimental duration (yr) in boreal forests (a & d), temperate forests (b & e) as well as subtropical and tropical forests (c & f).



**Fig. 5.** Meta-regression plots showing variations of effect size with MAT (°C), MAP (mm) and soil pH in boreal forests (a, d, g), temperate forests (b, e, h) as well as subtropical and tropical forests (c, f, i).

$$w_i = \frac{1}{v_i + \tau^2} \quad (3)$$

For easier interpretation of the results, the mean effect size was further transformed back to the relative change (%) in soil CH<sub>4</sub> uptake in response to N additions according to equation (4),

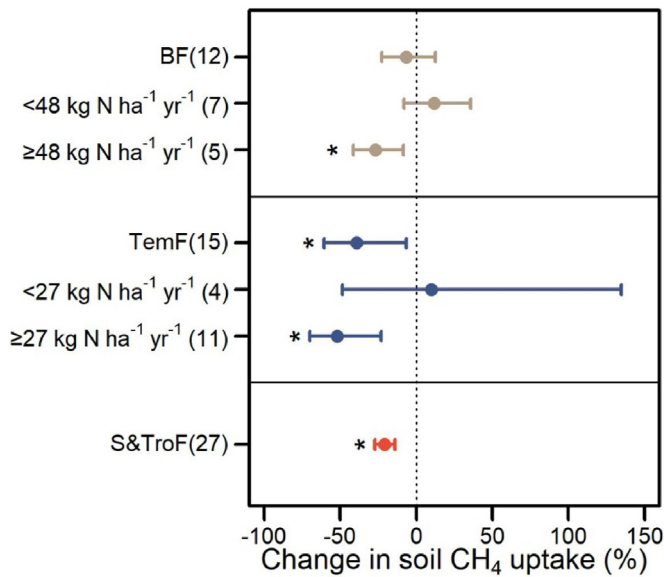
$$\text{relative change} = \left( \exp^{\text{effect size}} - 1 \right) \times 100\% \quad (4)$$

We also conducted meta-regression analyses for each forest biome to explore the relationship between effect size and multiple moderators, including N addition rate, the experimental duration, MAT, MAP, and soil pH. We tested whether there was a shift from a positive effect to a negative effect with increasing N addition rate. If such a shift occurred, we then identified the N threshold ( $N_0$ ) and estimated the mean effects of N additions below ( $<N_0$ ) and above the threshold level ( $\geq N_0$ ). The meta-analysis was conducted using

the *metafor* package (Viechtbauer, 2010). All statistical analyses were performed using R software (version 3.5.3; R Development Core Team, 2019; <http://www.r-project.org/>) with a significance level of  $p < 0.05$ .

### 3. Results

Soils were significant CH<sub>4</sub> sinks across boreal (growing-season mean =  $-0.069 \pm 0.019$  mg CH<sub>4</sub> m<sup>-2</sup> h<sup>-1</sup>,  $n = 7$ ), temperate (growing-season mean =  $-0.035 \pm 0.017$  mg CH<sub>4</sub> m<sup>-2</sup> h<sup>-1</sup>,  $n = 8$ ) as well as subtropical and tropical forests (annual mean =  $-0.020 \pm 0.004$  mg CH<sub>4</sub> m<sup>-2</sup> h<sup>-1</sup>,  $n = 17$ ) (Fig. 2). When combining all N addition trials, soil CH<sub>4</sub> uptake showed an overall decrease by 25% ( $p < 0.001$ ) in response to N additions in global forests (Fig. 3). However, the effect of N addition varied across forest biomes (Fig. 3). In boreal forest, N addition on average decreased soil CH<sub>4</sub> uptake by 7%, but the effect was statistically insignificant



**Fig. 6.** Relative changes (%) of soil CH<sub>4</sub> uptake in response to N additions in boreal (BF), temperate (TemF) as well as subtropical and tropical forests (S&TroF) accounting for differences in the N addition range. The effects of N additions below and above the threshold N levels ( $N_0$ ) are shown for boreal ( $N_0 = 48 \text{ kg N ha}^{-1} \text{ yr}^{-1}$ ) and temperate forests ( $N_0 = 27 \text{ kg N ha}^{-1} \text{ yr}^{-1}$ ), respectively. The asterisk (\*) indicates a significant difference from zero ( $p < 0.05$ ).

( $p > 0.1$ ). In contrast, N addition significantly decreased soil CH<sub>4</sub> uptake by 39% in temperate forest ( $p < 0.05$ ) and by 21% in subtropical and tropical forest ( $p < 0.001$ ).

Meta-regression analyses indicate a shift from a positive to a negative effect on soil CH<sub>4</sub> uptake with increasing N additions in boreal (threshold =  $48 \text{ kg N ha}^{-1} \text{ yr}^{-1}$ ) (Fig. 4a) and temperate forests (threshold =  $27 \text{ kg N ha}^{-1} \text{ yr}^{-1}$ ) (Fig. 4b), while no such shift was found in subtropical and tropical forests (Fig. 4c). Experimental duration generally exerted no significant influence on the effect size of N addition across all forest biomes (Fig. 4d–f). We further tested the roles of climate and soil moderators in each forest biome and found no significant influences of MAT, MAP and soil pH on the effect size of N addition (Fig. 5).

Considering the biome specific and dose dependent effect, we made a distinction of N addition below and above the thresholds in boreal and temperate forests, respectively (Fig. 6). The results show that N additions below the threshold  $48 \text{ kg N ha}^{-1} \text{ yr}^{-1}$  insignificantly increased soil CH<sub>4</sub> uptake by 12% in boreal forest ( $p > 0.1$ ), while N additions above the threshold significantly decreased soil CH<sub>4</sub> uptake by 27% ( $p < 0.01$ ). In temperate forest, N additions below the threshold  $27 \text{ kg N ha}^{-1} \text{ yr}^{-1}$  insignificantly increased soil CH<sub>4</sub> uptake by 10% ( $p > 0.1$ ), while N additions above the threshold significantly decreased soil CH<sub>4</sub> uptake by 52% ( $p < 0.01$ ).

#### 4. Discussion

Our results support the hypotheses of a shift from a positive to negative effect on soil CH<sub>4</sub> uptake with increasing N additions in relatively N-limited forest biomes. In boreal and temperate forests that are limited in N availability (Du and De Vries, 2018; Du et al., 2020b), low-level N addition generally stimulates forest growth (Schulte-Uebbing and De Vries, 2018) and further increase soil CH<sub>4</sub> oxidation via increasing evapotranspiration. In addition, low-level N addition can partially release N limitation of methanotrophic microorganisms and favors soil CH<sub>4</sub> uptake (Bodelier et al., 2000; Bodelier and Laanbroek, 2004). In contrast, high-level N addition

exerted a significant reduction of soil CH<sub>4</sub> uptake, being likely attributable to a direct inhibition of CH<sub>4</sub> oxidation by increased inorganic N concentrations (Schnell and King, 1994; Sitaula et al., 1995; Bodelier and Laanbroek, 2004) and/or an indirect effect due to soil acidification and an imbalance of N and phosphorus (Veraart et al., 2015; Zhang et al., 2020). Moreover, our results suggest a higher threshold of N addition for the shift from a positive to a negative effect in boreal forest than temperate forest (48 vs.  $27 \text{ kg N ha}^{-1} \text{ yr}^{-1}$ ). This is likely due to stronger limitation of soil N availability in boreal forest than temperate forest (Du and De Vries, 2018; Du et al., 2020b).

In view of the fact that current N deposition to boreal (mean =  $1.2 \text{ kg N ha}^{-1} \text{ yr}^{-1}$ ) and temperate forests (mean =  $7.3 \text{ kg N ha}^{-1} \text{ yr}^{-1}$ ) is generally below  $25 \text{ kg N ha}^{-1} \text{ yr}^{-1}$ , except part of temperate regions in China (Schwede et al., 2018), actual N deposition likely exerts a positive to neutral effect on soil CH<sub>4</sub> uptake in most of these forests (Fig. 6). Previous meta-analyses, not differentiating the biome specific and dose dependent effect of N addition estimated either a significant negative effect of N addition on soil CH<sub>4</sub> uptake across all forest biomes (Liu and Greaver, 2009) or an overall neutral effect (Aronson and Helliker, 2010). A recent literature review indicates that forest soil CH<sub>4</sub> sink in northern hemisphere has significantly declined during the past three decades and this trend has been attributed to an increase of precipitation (Ni and Groffman, 2018). Our analysis implies that N deposition is not likely to contribute to the reduction of this soil CH<sub>4</sub> sink. Compared with existing studies (e.g., Liu and Greaver, 2009; Aronson and Helliker, 2010), this work provides new insights and improves current understanding of biome specific and dose dependent effect of N addition on forest soil CH<sub>4</sub> sinks.

Our findings have important implications for improving the performance of process-based models to simulate and quantify the effect of global N deposition on soil CH<sub>4</sub> sinks (e.g., Ridgwell et al., 1999; Curry, 2007; Murguía-Flores et al., 2018). Currently, the existing models simply account for a negative effect of N inputs on soil CH<sub>4</sub> uptake by involving an inhibition factor (Ridgwell et al., 1999; Curry, 2007; Murguía-Flores et al., 2018). Using a process-based model, Murguía-Flores et al. (2018) estimated that N inputs via atmospheric deposition reduced global soil uptake of atmospheric CH<sub>4</sub> by  $1.38 \text{ Tg yr}^{-1}$ . However, our analysis indicates that N addition below a certain threshold level can exert a positive effect on soil CH<sub>4</sub> uptake in boreal and temperate forests, implying that existing models most likely overestimate the negative effect of N deposition on forest soil CH<sub>4</sub> sinks, especially in the boreal and temperate regions. This indicates the necessity of models to include a biome specific and dose-dependent N effect on soil CH<sub>4</sub> uptake across forest biomes.

Uncertainties might remain in our analysis due to relatively limited number of N addition experiments, especially in tropical forests ( $n = 4$ ). Moreover, the existing experimental studies are located unevenly in space. For instance, our search of the literature found few reports on the effects of N addition on soil CH<sub>4</sub> flux in the southern hemisphere (Fig. 1). In addition, experimental results with low levels of N addition (e.g.,  $< 30 \text{ kg N ha}^{-1} \text{ yr}^{-1}$ ) were relatively scarce and shed limited insights on the actual N deposition effect in forest ecosystems, most of which received relatively low levels of N deposition (Schwede et al., 2018). Nevertheless, our results represent the best knowledge we can derive at this stage because the current database is the most updated. To better understand the dynamics of forest soil CH<sub>4</sub> sinks in response to N deposition, we call for more field experiments using low levels of N addition.

#### 5. Conclusions

Our synthesis of experimental results indicates biome specific

and dose dependent effects of N addition on soil CH<sub>4</sub> uptake in global forests. In boreal and temperate forests, we found a shift from a positive to a negative effect on soil CH<sub>4</sub> uptake with increasing N additions, while N addition consistently decreased soil CH<sub>4</sub> uptake in subtropical and tropical forests. The thresholds for boreal and temperate forests (48 and 27 kg N ha<sup>-1</sup> yr<sup>-1</sup>) are much higher than the levels of N deposition in these forest biomes, except for some regions in Asia, implying that current understanding that N deposition reduces forest soil CH<sub>4</sub> uptake is flawed by high N addition rates. Compared with existing meta-analyses (Liu and Greaver, 2009; Aronson and Helliker, 2010), our work improves current understanding of actual N deposition effect on soil CH<sub>4</sub> sinks in global forests.

## Author statement

E.D. conceived the study. N.X. and E.D. compiled the database and analyzed the data. All authors wrote and revised the manuscript.

## Declaration of competing interest

The authors declare no conflict of interest.

## Acknowledgement

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## Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.envpol.2020.114751>.

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